Capstone Project: Loan Default Prediction

Final Submission Report

Practical Data Science track (Classification)

MIT Professional Education - Applied AI and Data Science Program

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1. Executive Summary

A bank's consumer credit department asked for an empirically derived, interpretable model that predicts which home equity loan applicants are likely to default, together with guidance on which applicant attributes should drive approval decisions. Using the Home Equity (HMEQ) dataset of **5,960 recent loans**, of which **19.9% defaulted**, this project explored the data, built and tuned linear and tree-based classifiers, compared them on a business-driven metric and arrived at a deployable solution design.

Most important findings from the analysis

- **Missing information is the strongest single risk signal.** Applicants with no recorded debt-to-income ratio defaulted 62% of the time versus 8.6% when the ratio was recorded. Applicants with no property valuation defaulted 94% of the time. Unverifiable income or collateral should be treated as a red flag, not a neutral gap.
- **Debt-to-income has a hard threshold.** Between a 20% and a 40% ratio only 4% to 7% of applicants default; between 40% and 45% the rate is 11%; above 45% it is 87% to 100%. A 45% cut-off is a simple, defensible policy rule.
- **Past repayment behaviour predicts future behaviour.** The default rate rises from 17% with no major derogatory report to 39% with one, 51% with two and over 70% with three or more. Delinquent credit lines show the same pattern (14% at zero, 34% at one, 100% at six or more), as do recent credit inquiries.
- **A long credit history protects.** Applicants whose oldest credit line is under five years old default 44% of the time; above twenty years the rate is about 10%.
- **Small loans are not safe loans.** Loans under $5,000 default 57% of the time, almost three times the average.
- **Occupation and loan purpose matter little on their own.** Sales and self-employed applicants default more often (35% and 30%) and office workers least (13%). At the conventional decision threshold, removing occupation, loan purpose and nine minor flags from the forest changes its performance by about a point; at the cost-optimal threshold removing occupation alone raises the expected cost of credit errors by about 10%, so occupation is kept subject to the fair-lending review, with a 21-input model without it as the fallback.
- **Tree-based models beat the linear model.** The relationships in this data are thresholds and cliffs rather than straight lines. The Random Forest scored 0.96 on risk-ranking quality (ROC-AUC, where

1.0 means every defaulter is ranked above every repayer) against 0.91 for logistic regression, and at a similar share of defaulters caught it declined far fewer good customers. The final model's configuration (class weights, tree size, input set) and its decision threshold were chosen together on the training data alone, by the expected dollar cost of their mistakes on training applicants the model had not seen, under a rule fixed in advance; performance was then measured once on the held-out applicants.

Final proposed model specification

Element	Specification
Scoring engine	Random Forest classifier, 200 fully grown trees, no class weights (giving defaulters extra weight in training made no measurable difference once the decision threshold was chosen on cost), standard settings otherwise
Inputs	All 27 model inputs, in the order the pipeline produces them: the 10 financial attributes (LOAN, MORTDUE, VALUE, YOJ, DEROG, DELINQ, CLAGE, NINQ, CLNO, DEBTINC), a 0/1 missing-value flag for each of the 11 fields with gaps (MORTDUE, VALUE, REASON, JOB, YOJ, DEROG, DELINQ, CLAGE, NINQ, CLNO, DEBTINC) and six 0/1 indicators from one-hot encoding (REASON = HomeImp; JOB = Office, Other, ProfExe, Sales, Self). Fallback if the fair-lending review requires it: the same forest without any occupation information (21 inputs: the five occupation indicators and the occupation-missing flag removed), about 10% more expected credit-error cost. A 12-input lean model (the ten attributes plus the DEBTINC and VALUE missing flags) was also tested and costs about a fifth more.
Pre-processing	Median imputation for numerical gaps and mode for categorical gaps, both learned on the training rows; a missing-value flag for every imputed field; no outlier capping for the tree model
Decision threshold	Flag as likely default when predicted probability > 0.15. The threshold minimises the indicative expected cost of credit errors on repeated out-of-fold predictions of the training rows; it is unchanged for cost ratios between 3:1 and 6:1, on a finer grid and on two other train/test splits, and sits just below the textbook break-even for the stated costs ($2,000 / $11,000 = 0.18) because the forest's scores understate risk in the decision range (applicants scored 0.15 to 0.20 defaulted about 23% of the time).
Explanation layer	A depth-4 surrogate decision tree fitted to the forest's out-of-fold decisions, which reproduces 93% of them on unseen applicants (against 73% for never flagging); its three rules (debt-to-income ratio missing; ratio above about 44%; at or below 44%, a delinquent credit line together with a major derogatory report) supply the adverse-action reasons required under the Equal Credit Opportunity Act, complemented by per-applicant top risk drivers
Measure of success	Recall on the defaulter class (share of future defaulters caught), with precision and F1 as guard-rails against declining too many good customers; the final model and its threshold were then chosen on the expected cost of the two errors, so precision is priced in explicitly
Performance on unseen applicants	92.7% of defaulters caught (recall), 68.1% precision, 89.9% accuracy, risk-ranking quality (ROC-AUC) 0.96. At the default 0.5 threshold the same model gives 68.6% recall, 84.2% precision and 91.2% accuracy.
Validation	70/30 stratified split held out before any pre-processing statistic was fitted; grid searches by 3- and 5-fold stratified cross-validation; the final model's settings and its threshold selected by the expected cost of mistakes on training applicants the model had not seen (three rounds of 5-fold cross-validation, 15 validation folds) under a rule fixed in advance, with robustness checks (finer threshold grid, two other train/test splits, calibration table); the test set scored once for performance (permutation importance is also computed on it, for interpretation only)

Key next steps

- **Decide and confirm the threshold:** the head of consumer credit approves the three-band policy (approve / refer / decline); finance supplies actual loss-given-default and margin figures so the 0.15 threshold can be confirmed.
- **Prepare the process:** make income and property-valuation verification mandatory fields, since their absence is the strongest default signal; build the scoring pipeline with reason codes.
- **Prove it safely:** run the model in shadow mode alongside manual decisions for one to two months and complete a fair-lending review before switching it on.
- **Make the most of it:** automate clear approvals and declines, route the uncertain band to underwriters, monitor recall, precision and approval rate monthly, and re-train yearly.
- **Improve it:** add reject inference, loan-to-value and payment-to-income features, probability calibration and time-based validation in the next iteration.

Per 1,000 applicants the proposed model is expected to catch about 185 of the 200 defaulters, miss about 15, and decline about 87 customers who would have repaid. Under indicative cost assumptions (a missed defaulter costs about $9,000, a wrongly declined good customer about $2,000) this cuts default-related cost from roughly **$1.8 million per 1,000 applicants** today to about **$0.3 million**, a reduction of about 83%.

2. Problem and Solution Summary

Summary of the problem

A large share of a retail bank's profit comes from interest on home loans and home equity lines of credit, and the single biggest threat to that profit is the borrower who does not repay: one non-performing loan wipes out the interest earned on many good ones. Today the approval decision is made by an underwriter's manual review of each application. The process is slow, costly, inconsistent across underwriters and vulnerable to human error and bias. Earlier attempts to automate it relied on hand-written heuristics.

The consumer credit department wants to replace that process with an empirically derived, statistically sound credit-scoring model built from the bank's own data on recent applicants, in line with the Equal Credit Opportunity Act. The model must do two things at once: predict at application time which applicants are likely to default, and remain interpretable enough that the bank can give a specific, defensible reason for every rejection. Formally this is a supervised binary classification problem on 5,960 historical loans with 12 attributes known at application time and a target indicating default or serious delinquency.

Why this solution design

The metric follows the money. The two errors a scoring model can make do not cost the same. Approving someone who then defaults (a missed defaulter) costs the bank most of the principal plus collection costs; declining someone who would have repaid costs only the interest margin on one customer. The primary measure of success is therefore the share of future defaulters the model catches (recall), with precision and F1 tracked so the model does not simply decline everyone.

Missing values were treated as information, not noise. Because the absence of a debt-to-income ratio or a property valuation was itself the strongest predictor of default, every imputed field received a 'was

missing' indicator before imputation. This single step is responsible for a large part of every model's performance.

Linear and non-linear techniques were compared head to head. Logistic regression (the transparent linear benchmark), a decision tree (readable rules), a bagged ensemble of trees and a random forest were trained on the same 70/30 stratified split, tuned with cross-validated grid search optimising recall, and evaluated on the held-out 30%. Building up from one tree to bagging to a forest makes the contribution of each idea visible. Time-series techniques were not applicable because the applications carry no dates. The table below summarises the result.

Model (held-out test set, 1,788 applicants)	Defaulters caught	Precision	Accuracy	Risk-ranking quality (ROC-AUC)
Logistic Regression (plain)	57%	77%	88%	0.91
Logistic Regression (balanced class weights)	80%	61%	86%	0.91
Decision Tree (fully grown)	69%	71%	88%	0.81
Decision Tree (tuned, pruned, balanced)	82%	53%	82%	0.88
Bagged trees (200 trees)	68%	83%	91%	0.95
Random Forest (default; the proposed model at the 0.5 threshold)	69%	84%	91%	0.96
Random Forest (class weights 0.2/0.8)	77%	77%	91%	0.96
Random Forest (tuned for recall)	79%	66%	88%	0.93
Random Forest (tuned, 12 features only)	78%	66%	88%	0.93
Proposed: Random Forest, threshold 0.15	93%	68%	90%	0.96

Table 1. Performance of the techniques explored on the held-out test set (1,788 applicants), at the conventional 0.5 threshold except for the proposed model.

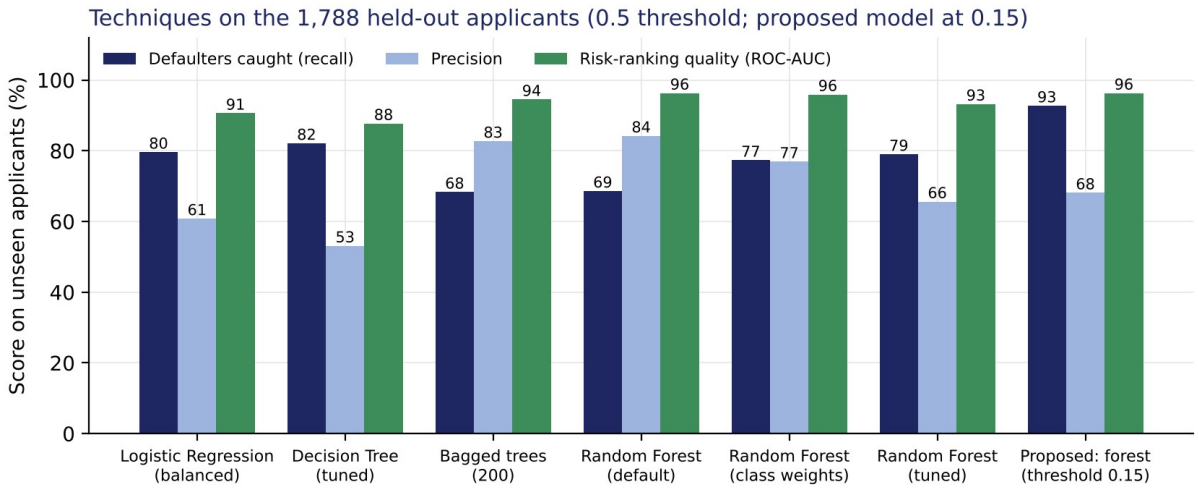


Figure 1. Share of defaulters caught, precision and risk-ranking quality of each technique on the held-out applicants.

The Random Forest was chosen as the scoring engine because it offers the best balance on the chosen metric. The tuned decision tree catches slightly more defaulters at the 0.5 threshold (82%) but declines almost one good customer for every defaulter it catches (53% precision). The balanced logistic regression catches 80% but with 61% precision, because a straight line cannot represent the threshold effects in the

data. Bagging 200 trees shows the value of averaging: at the same recall as a single fully grown tree, precision rises from 71% to 83% and ROC-AUC from 0.81 to 0.95. The forests add feature sub-sampling: with class weights the forest catches 77% at 77% precision at the 0.5 threshold, and both forests rank applicants by risk most accurately of all models (ROC-AUC 0.96), which is what makes the next step, moving the decision threshold, possible.

The final model and its threshold were chosen on the training data, from business cost rather than convention. Two settings decide how readily a forest flags an applicant: the extra weight given to defaulters during training (class weights) and the probability cut-off. They were chosen together, on the training data alone, by one yardstick: the expected dollar cost of the model's mistakes, estimated on training applicants the model had not seen (three rounds of 5-fold cross-validation). Fifteen forest set-ups (five weightings, each with fully grown trees or trees limited to at least 5 or 10 applicants per leaf) and then five final candidates (the best set-up on all 27 inputs, without occupation, and on a 12-input subset, plus the recall-tuned forest on 27 and 12 inputs) were each scored at their own cost-minimising cut-off, under a rule fixed beforehand: lowest average cost wins, and when several candidates are within the run-to-run noise of the estimate (about $10,000 per 1,000 applicants) the simplest wins. Fully grown trees beat limited trees for every weighting. Among them the weightings made no measurable difference (the unweighted, 0.3/0.7 and 0.2/0.8 forests were within that noise of each other, at cut-offs of 0.15, 0.20 and 0.25), so the unweighted forest on all 27 inputs was selected; the recall-tuned grid winner cost about half again as much once precision is priced in, the forest without occupation about 10% more and the 12-input forest about a fifth more. Scored the same way at their own best cut-offs, the balanced logistic regression and the recall-tuned decision tree cost about $550,000 and $607,000 per 1,000 applicants against $317,000 for the forest, so the forest wins on the chosen yardstick as well as on ranking quality. The cost-minimising threshold is 0.15, unchanged for any cost ratio between 3:1 and 6:1, on a finer grid and on two other train/test splits; it sits just below the textbook break-even of 0.18 for a perfectly calibrated score because the forest's scores, which rank applicants well, understate risk in the decision range (applicants scored 0.15 to 0.20 defaulted about 23% of the time). Scored once on the held-out applicants at that threshold, the forest catches 93% of defaulters at 68% precision, against 69% and 84% at the conventional 0.5. The test result (cost about $304,000 ± $28,000 per 1,000 applicants) is consistent with the out-of-fold estimate ($317,000); the choice between the closely matched configurations rests on the out-of-fold comparison, which a single test evaluation is too noisy to adjudicate.

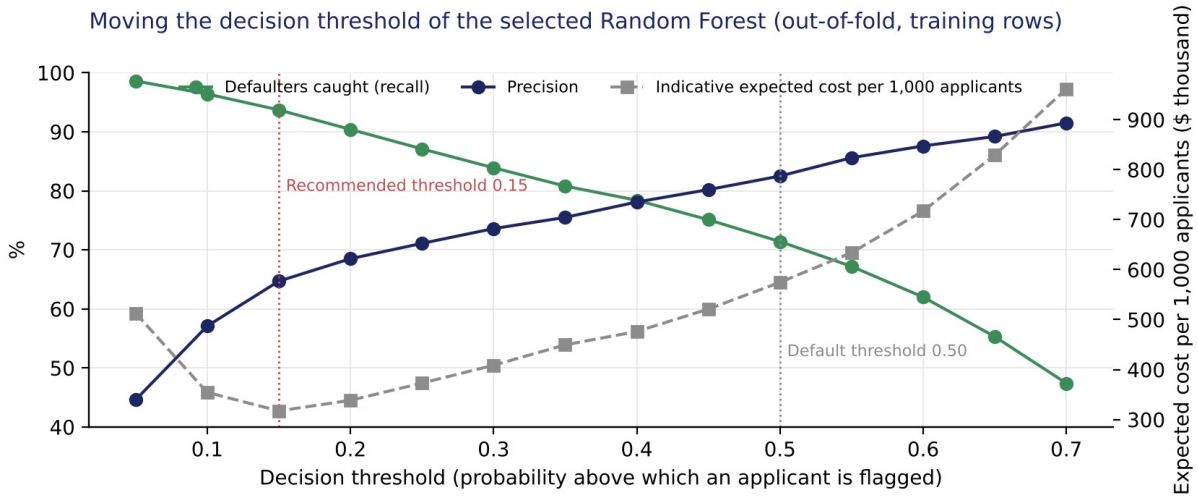


Figure 2. Recall, precision and indicative expected cost per 1,000 applicants as the forest's decision threshold moves, measured on out-of-fold predictions of the training rows (the test set played no part in the choice).

Interpretability is provided by a surrogate decision tree that follows the forest. A forest of 200 trees cannot be read as a rule list, so a four-level decision tree was fitted to the forest's own decisions (flag or clear at the 0.15 threshold, taken from the out-of-fold scores so that the tree learns how the forest treats applicants it has not seen) rather than to the original labels. It reproduces 93% of the forest's decisions on unseen applicants, against 73% for a rule that never flags anyone, and its rules are the same drivers the analysis found throughout: is the debt-to-income ratio missing (then flag); if recorded, is it above about 44% (then flag); at or below 44%, is there a delinquent credit line together with a major derogatory report (then flag, otherwise clear). Those rules supply the adverse-action reasons; the forest's feature importances (Figure 3) and per-applicant top risk drivers document the score itself.

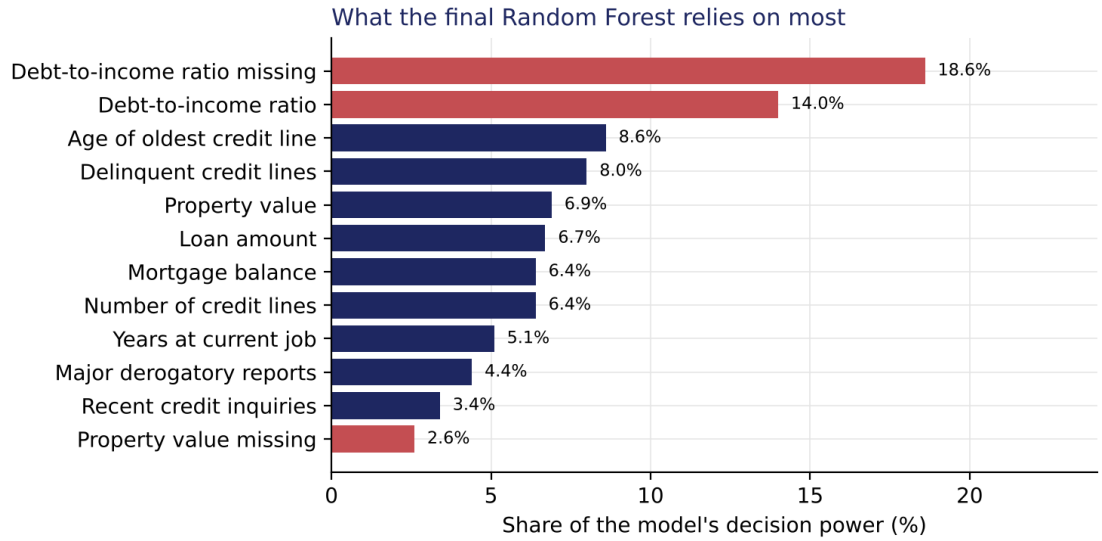


Figure 3. The twelve inputs the final Random Forest relies on most (impurity-based importance). The debt-to-income ratio, its missing-value flag, the number of delinquent credit lines and the age of the credit history together carry about half of the model's decision power; permutation importance ranks the same four at the top.

How the solution affects the business

- **Fewer bad loans.** Of every 200 defaulters in 1,000 applications the model flags about 185 before the money leaves the bank, against none today.
- **Protected good business.** About 87 of every 1,000 applicants would be declined or referred although they would have repaid. This is the price of catching defaulters; it is far lower than the single decision tree's 145 and can be reduced further by routing borderline cases to a human underwriter instead of declining them outright. Stated as rates: the model flags about 27% of all applicants (an approval rate of about 73% on a population the bank approves in full today, 84% at the conventional 0.5 cut-off), and about 11% of creditworthy applicants are wrongly flagged at 0.15 against about 3% at 0.5. Every flag that ends in a decline needs an adverse-action notice, which is one more reason to route the uncertain middle to a person.
- **Faster, more consistent decisions.** About seven in ten applications fall in the clear-approval band and about a quarter in the clear-decline band, so both can be automated; underwriters focus on the roughly 7% in the uncertain band around the threshold.
- **Defensible decisions.** Every rejection maps to a small set of plain-language reasons (debt burden, unverified income, delinquencies, derogatory reports, short credit history) that match both the data and lending common sense.

3. Recommendations for Implementation

Key recommendations

1. **Deploy the Random Forest as a decision-support score,** integrated into the existing underwriting workflow, with the probability threshold of 0.15 as the initial setting and a three-band outcome: approve (predicted default probability below about 0.10), refer to an underwriter (0.10 to 0.25) and decline (above 0.25). On the held-out applicants these bands hold about 69%, 7% and 24% of applications, with observed default rates of about 1%, 24% and 74%, so the bands keep a human in the loop exactly where the model is least certain without creating a large manual workload. The edges are round numbers around the cut-off, set before looking at the test set roughly where the expected cost rises 10% to 20% above its minimum; tune them to underwriting capacity in the shadow period.
2. **Make income and collateral verification mandatory before scoring.** A missing debt-to-income ratio or property valuation is the strongest default signal in the data. Requiring these fields removes the ambiguity, improves the data for the model and is itself a prudent lending control. Sequence it with the shadow period: the current model relies most on the missing-ratio signal, so once verification is mandatory that group shrinks and the model must be re-trained on the verified population before go-live; on applicants with a recorded ratio it already catches 83% of defaulters at 80% precision while flagging only 9%.
3. **Adopt the explicit policy rules the analysis surfaced.** Decline or escalate applications with a debt-to-income ratio above 45% (87% to 100% of them defaulted), and refer to an underwriter applications with two or more delinquent credit lines or two or more major derogatory reports (observed default rates of 45% and 51% and above). These are escalation rules read directly from the observed default rates, applied before the score; they are transparent and easy to communicate and use the same drivers as the model, which applies them more finely (its surrogate flags a delinquent line together with a derogatory report at any recorded ratio). Where a hard rule and the score disagree, the rule takes precedence.
4. **Use the surrogate decision tree for adverse-action reasons.** Generate the reason codes for every decline from the surrogate tree's rules (which reproduce 93% of the forest's flag decisions at the 0.15 cut-off) and the applicant's top risk drivers from the forest, and store them with the application record for audit and for responses to applicants and regulators. Because automatic declines are issued above 0.25, the forest's own top risk drivers are the primary reason for those, with the surrogate rule as the plain-language summary; a referral that ends in a decline carries the underwriter's reason.
5. **Confirm the threshold with real cost figures.** Replace the indicative $9,000 and $2,000 assumptions with the bank's actual loss-given-default and customer lifetime margin, and re-derive the threshold on out-of-fold predictions before go-live. The analysis shows the optimum stays at 0.15 for any cost ratio between 3:1 and 6:1 and would move to 0.20 if the ratio were nearer 2:1, so this step refines the operating point rather than the design.
6. **Run a shadow period and a fairness review before switching on.** Score every application alongside the manual decision for one to two months, including applications the current process declines (the training data contains only approved loans), compare outcomes, and test approval and error rates across protected groups. If the review requires occupation to be excluded, switch to the 21-input fallback model without occupation, at about 10% more expected credit-error cost.
7. **Monitor and re-train.** Track recall, precision, approval rate (about 73% expected on today's approved population at the 0.15 cut-off, with about 69% approved automatically, 7% referred and

24% declined under the bands) and the share of applications with missing fields monthly; re-train at least yearly or when the default rate or input distributions drift.

Key actionables for stakeholders

Stakeholder	Actionables
Head of Consumer Credit (sponsor)	Approve the three-band decision policy and the initial 0.15 threshold; own the risk-appetite decision the threshold represents; sign off the go-live after the shadow period.
Credit risk and finance	Supply actual loss-given-default and margin figures to confirm the threshold; validate the cost-benefit case; define the monitoring thresholds that trigger re-training.
Underwriting operations	Make income and valuation verification mandatory fields; redesign the workflow around approve / refer / decline bands; train underwriters on reading reason codes and handling referred cases.
Data and technology	Build the scoring pipeline (imputation, missing flags, model, threshold, reason codes); integrate with the loan-origination system; set up logging, monitoring dashboards and a yearly re-training job.
Compliance and legal	Review adverse-action reason codes against ECOA requirements; run the disparate-impact analysis and decide whether occupation may stay in the model; approve model-risk documentation.
Data science	Maintain the model, run the shadow comparison, extend the analysis (see further analysis below) and report monthly performance.

Expected benefits and costs

The figures below are indicative and rest on the stated assumptions; they are meant to size the opportunity and to be refined with the bank's own numbers.

Assumptions. Application volume of 6,000 per year (the size of the dataset). Defaulted loans averaged $16,900 in the data; with about half recovered through the collateral, a missed defaulter costs about $9,000. A wrongly declined good customer costs about $2,000 in forgone net interest margin and acquisition spend (illustratively, a net margin of about three percentage points on a $16,000 to $17,000 balance over three to four years plus the cost of acquiring a replacement customer); finance should replace both figures, and if the true ratio were nearer 2:1 the cost-minimising cut-off would move to 0.20 (about 90% recall and 83 good customers declined per 1,000 on out-of-fold data). Model performance in production equals test-set performance (93% of defaulters caught, 68% precision at threshold 0.15), on an applicant population like the one the current process approves: the benefit is an additional screen on applicants the current process approves, and the outcomes of applicants the manual process declines today are unknown until the shadow period. It is also a before-verification estimate: two thirds of defaulters sit in the fifth of applicants with no recorded debt-to-income ratio, a group the model flags almost entirely (96%) and that mandatory verification will shrink; on applicants with a recorded ratio the model catches 83% of defaulters at 80% precision while flagging 9%, the better guide to its behaviour once verification is in place and the model is re-trained. All figures are for the single 0.15 cut-off; the approve / refer / decline bands are the operating procedure around it, and whatever underwriters approve inside the referral band (about 7% of applications, one in four of whom defaulted) trades some of the modelled benefit for a human check: under the bands the share of defaulters caught lies between about 87% (if every referral were approved) and 96% (if every referral were declined), and 93% is the single-cut figure.

Item (per year, 6,000 applications)	Current process	With the proposed model
Loans that default	about 1,200 (20%)	about 90 slip through (15 per 1,000)
Default-related losses	about $10.8 million	about $0.8 million
Good customers declined or referred	none attributable to a model	about 520 (87 per 1,000), margin forgone about $1.0 million
Total indicative cost of credit errors	about $10.8 million	about $1.8 million
Indicative annual benefit		about $9.0 million gross; about $8.9 million after running costs; about $8.5 million in the first year after the one-off implementation cost

Table 2. Indicative annual benefit of the proposed solution under the stated assumptions.

Costs. A one-off implementation of roughly $300,000 to $400,000 (data pipeline and integration, validation, fairness review, documentation and training) and running costs of roughly $100,000 per year (monitoring, re-training, model-risk reviews). Even if the realised benefit were only a quarter of the indicative figure, the solution would pay back within the first year.

Non-financial benefits. Faster decisions for clear cases, consistent treatment of applicants, documented and auditable reasons for every decline, and a data asset (verified income and valuation on every application) that improves every future model.

Potential risks and challenges

Risk or challenge	Why it matters	Mitigation
Selection bias in the training data	The data covers only loans the bank approved; applicants declined under the old process are absent, so the model has not seen the full applicant population.	Shadow period on all applications; reject inference in the next iteration; conservative referral band.
The missing-value signal may change	Part of the 'ratio missing' signal reflects how the old process recorded data. Once verification is mandatory the feature will shift.	Introduce mandatory verification alongside the shadow period, re-train on the verified population before go-live and monitor the share of missing fields; on applicants with a recorded ratio the model already catches 83% of defaulters at 80% precision.
Economic drift	Default rates move with interest rates and employment; a model trained on one period can mis-calibrate in another.	Monthly monitoring of default rate and score distribution; yearly re-training; threshold review.
Fair-lending and proxy effects	Occupation, credit-history length (an age proxy) and the missing-ratio flag could act as proxies for protected characteristics, and the lower cut-off multiplies adverse actions against creditworthy applicants (about 11% of them at 0.15 against 3% at 0.5).	Disparate-impact testing of approval and error rates before launch, annually and whenever the cut-off or bands move; switch to the 21-input fallback (no occupation information, about 10% more expected credit-error cost) if the review requires it.
Interpretability of the ensemble	Regulators and applicants need specific reasons; a forest alone is hard to narrate.	Decision-tree explanation layer, per-applicant reason codes, model-risk documentation.

Risk or challenge	Why it matters	Mitigation
Over-reliance on the score	Automation bias could lead underwriters to stop exercising judgement on referred cases.	Keep the referral band, require documented overrides, audit override outcomes.
Data quality	Implausible values exist (debt-to-income above 100%, credit history of 97 years).	Input validation rules at capture; range checks in the scoring pipeline.

Further analysis and associated problems to solve

- **Reject inference:** obtain outcomes or proxies for applicants declined under the manual process so the model reflects the whole applicant population.
- **Feature engineering:** add a loan-to-value ratio ((LOAN + MORTDUE) / VALUE) and payment-to-income measures, which the trees currently approximate.
- **Alternative ensembles and calibration:** test gradient boosting and calibrate predicted probabilities so that a score of 0.15 means exactly a 15% chance of default (the unweighted forest's scores rank applicants well but understate risk in the decision range: applicants scored 0.15 to 0.20 defaulted about 23% of the time), which simplifies threshold setting and pricing.
- **Time-based validation:** once application dates are available, validate on later periods rather than a random split to measure drift realistically.
- **Risk-based pricing:** use the probability of default to set interest rates or loan limits for the referral band instead of a binary approve/decline.
- **Collections and early warning:** extend the model to existing loans, using payment behaviour after origination to flag accounts heading towards delinquency.
- **Per-applicant explanations:** add SHAP-style reason codes so each decision lists the attributes that moved the score most, in applicant-friendly language.

Appendix: Supporting Analysis

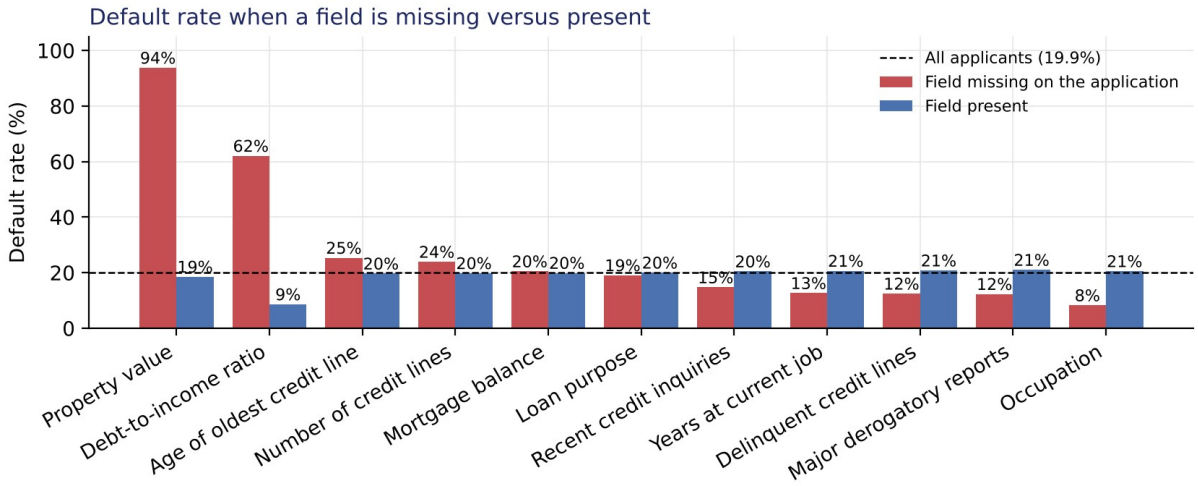


Figure A1. Default rate when an attribute is missing versus present. A missing debt-to-income ratio or property valuation is the strongest signal in the data.

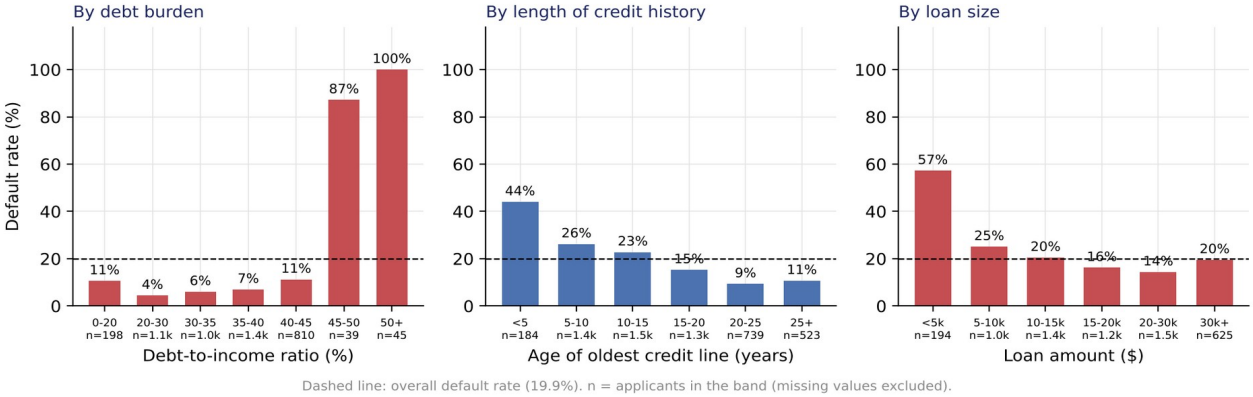


Figure A2. Default rate by debt-to-income band, age of oldest credit line and loan amount.

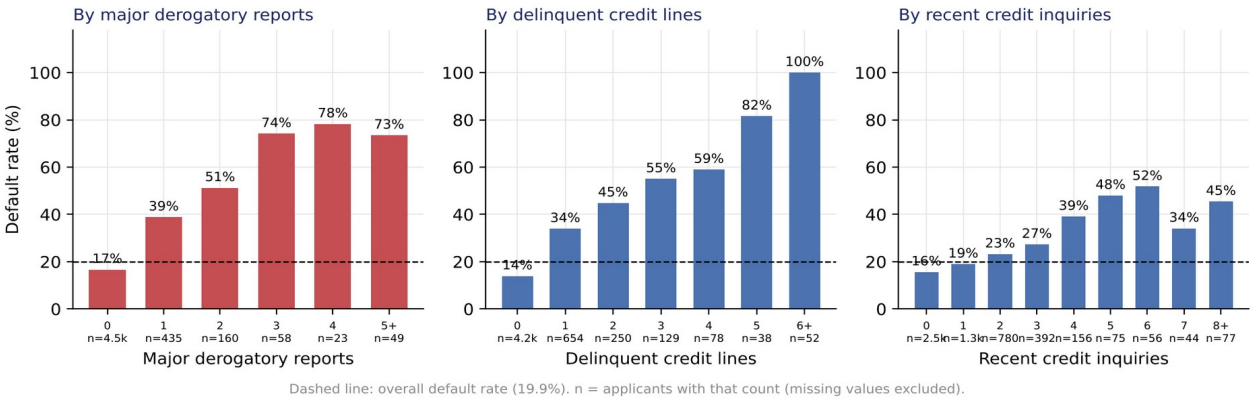


Figure A3. Default rate by number of derogatory reports, delinquent credit lines and recent credit inquiries.

Proposed model on the 1,788 held-out applicants	Model flags as likely default	Model clears as likely to repay	Total
Actually defaulted	331 (caught)	26 (missed)	357
Actually repaid	155 (wrongly flagged)	1,276 (correctly cleared)	1,431
Total	486	1,302	1,788

Table A1. Decisions of the proposed model (Random Forest, threshold 0.15) on the held-out applicants: 92.7% of defaulters caught; 68.1% of flagged applicants were real defaulters; 89.9% of all decisions correct.

The complete analysis, including all exploratory charts, model code, cross-validation results and the full decision-tree rules, is in the accompanying notebook *Loan_Default_Prediction_Capstone.html*.